
Findings Ledger: A Finding-by-Finding Analysis of FC–SC Connectome Translation

Each finding across both parcellations and all estimators

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Abstract

This document is the long-form companion to the main preprint. It walks every finding in the project ledger (F1–F10) one at a time, and for each finding shows the evidence *across both parcellations* (Glasser, 360 regions / 64,620 edges; 4S456Parcels, 456 regions / 103,740 edges) *and across all three estimators* (PCA→PLS, BayesianRidge, KernelRidge). Reconstruction leads with demeaned_pearson (the per-subject, group-mean-removed correlation); downstream cognition leads with BayesianRidge lift over the bv+demo baseline. Every figure is generated from committed CSV outputs. *Provenance*: F1–F5 and F9–F13 come from the frozen reproduction-grid CSVs (reproduction/outputs/) plus the noise/tractography/reduction sanity outputs; F6–F8 (family-structure and PC-mechanism analyses) come from a *separate* family-mechanism pipeline (reproduction/family_mechanism/outputs/) that reuses the same frozen family-aware splits but adds the sibling/twin task — which is not part of the three main grid files. Beyond the original ledger (F1–F10), three additional findings from this analysis (F11–F13) stress-test the affirmative claim with multiplicity correction, a Bayesian re-analysis, and a quantification of imputation harm and cross-modal loss.

Table 1: The findings ledger. “Status” is the current replication state across the two parcellations.

F	Meaning	Status
F1	FC→SC > SC→FC asymmetry	replicated both parcellations
F2	anatomy→SC / demographics→FC double dissociation	replicated both
F3	imputed connectomes are measurable, but utility is asymmetric	replicated both (via F5)
F4	observed FC carries a crystallized-cognition signal (narrow, FDR-surviving cell)	replicated both
F5	SC underperforms; imputation does not transfer utility; pred_FC harmful	replicated both
F6	predicted connectomes carry heritable family signal	replicated both
F7	reconstruction objective and identification objective trade off	replicated both
F8	exploratory FC-predictable low-variance SC mode (component index not stable)	partial / hypothesis-generating
F9	richer tractography representation does not help	Glasser / exploratory
F10	nonlinear models and more data do not break the ceiling	Glasser / exploratory
<i>Extra findings from this analysis (not in the original ledger):</i>		
F11	the observed-FC cognition lift is real but multiplicity-fragile	both, this analysis
F12	Bayesian corroboration (shrinkage / ROPE / Bayes factors)	both, this analysis
F13	imputation harm is direction-specific; cross-modal loss vs same-estimator oracle	both, this analysis

Note on estimators (read once; applies throughout). Reconstruction (connectome→connectome, a high-dimensional target) is reported for all three estimators. For *downstream scalar* targets (cognition/age, a 1-D output), the RBF KernelRidge rows are *a priori* ill-conditioned: regressing a single scalar through an RBF kernel under the project’s α/γ grid is numerically unstable (extreme negative R^2 , parcellation-inconsistent and sign-flipping lifts for parcellation-independent inputs). This is a principled, pre-declared exclusion of the *scalar* KR rows — not a post-hoc choice made whenever KR disagrees. BayesianRidge is therefore the reportable downstream estimator; PCA→PLS scalar rows agree with it directionally. Where KR scalar bars appear below (F4, F5, F13), they are shown only to make this instability visible, and are not interpreted. *Separately*, a float32 PCA conditioning bug in the scalar path (the bv+demo one-hots for sex/race are exactly collinear, so float32 SVD scrambled the demographic directions seed-dependently) was fixed by casting to float64 (commit 69add40); its *entire* blast radius was the sex/age leak diagnostics (now ≈ 1.0 as expected since bv+demo contains those labels). Cognition (F4/F5) is byte-identical and reconstruction (F1/F2/F13) unaffected.

1 F1 · Cross-modal prediction is directional (FC→SC > SC→FC)

Claim. FC reconstructs individual deviations in SC better than SC reconstructs FC.

Across parcellations (PCA→PLS, demeaned_pearson). Glasser: FC→SC 0.136 vs SC→FC 0.084, ratio $1.61\times$. 4S456: 0.146 vs 0.087, ratio $1.68\times$. Every one of the 10 frozen seeds exceeds a ratio of 1 in both atlases (Fig. 1D).

Across estimators. The ratio is essentially unchanged when the estimator is swapped (Fig. 1A–B): in Glasser it is $1.61\times$ (PCA→PLS), $\sim 1.6\times$ (BayesianRidge), and $1.65\times$ (KernelRidge); in 4S456 it runs $1.68\text{--}1.73\times$. The asymmetry is therefore not a property of one mapping.

Across reductions (Glasser). The ratio stays > 1 for full PLS on the raw 64,620-edge vectors ($1.81\times$), learned PCA ($1.62\times$), and three Johnson–Lindenstrauss random projections ($1.39\text{--}1.55\times$), all at Wilcoxon $p \leq 0.001$ (Fig. 1C). Full PLS (no reduction) gives an even *stronger* ratio, so the PCA step modestly attenuates rather than manufactures the effect.

Status. replicated both parcellations. The finer 4S456 atlas slightly raises every input→SC row, consistent with more predictable structural detail rather than added noise.

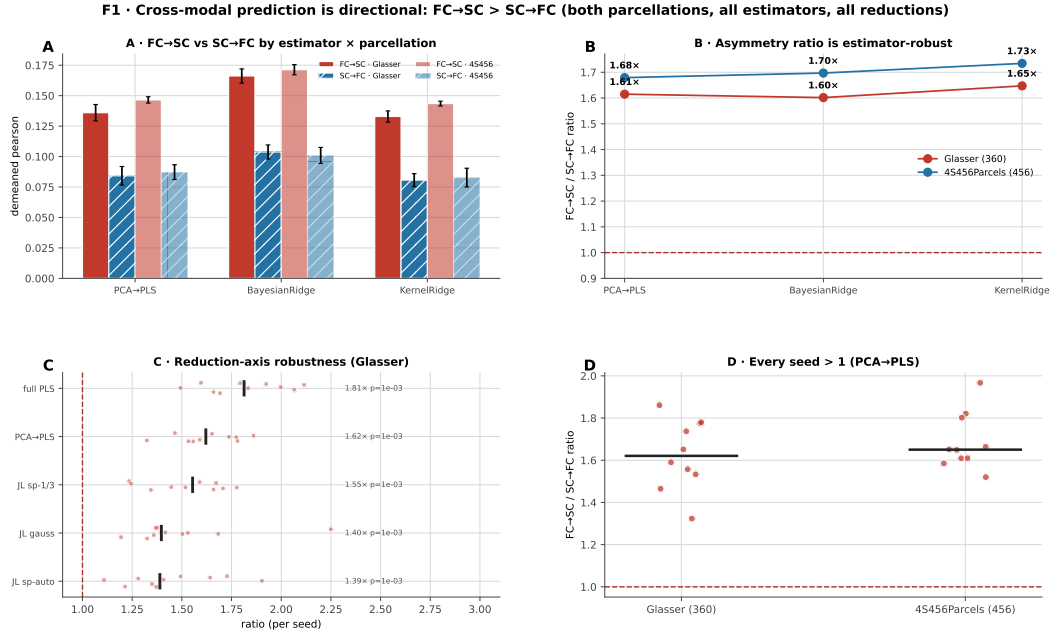


Figure 1: **F1.** (A) FC→SC vs SC→FC by estimator and parcellation; (B) the ratio is estimator-robust; (C) and survives every reduction axis; (D) every seed exceeds 1.

2 F2 · Anatomy→SC / demographics→FC double dissociation

Claim. “Subject information” is not one generic confound: brain-volume features predict SC better than demographics do, while demographics predict FC more than twice as well as brain volumes do.

Across parcellations (PCA→PLS, demeaned_pearson). Glasser: $bv \rightarrow SC$ 0.162 > $demo \rightarrow SC$ 0.130, but $demo \rightarrow FC$ 0.111 > $bv \rightarrow FC$ 0.049 ($a > 2 \times$ gap). The same crossover holds in 4S456 (Fig. 2A–B).

Across estimators. The dissociation index ($bv - demo$) is positive for $\rightarrow SC$ and negative for $\rightarrow FC$ under all three estimators and both parcellations (Fig. 2C), so the crossover is not estimator-specific.

Status. **replicated both parcellations.** *Caveat (see needed-experiments E2):* brain volume mechanically inflates streamline counts, so part of the $bv \rightarrow SC$ arm may be a tractography construction effect; a volume-normalized SC re-run is the clean test.

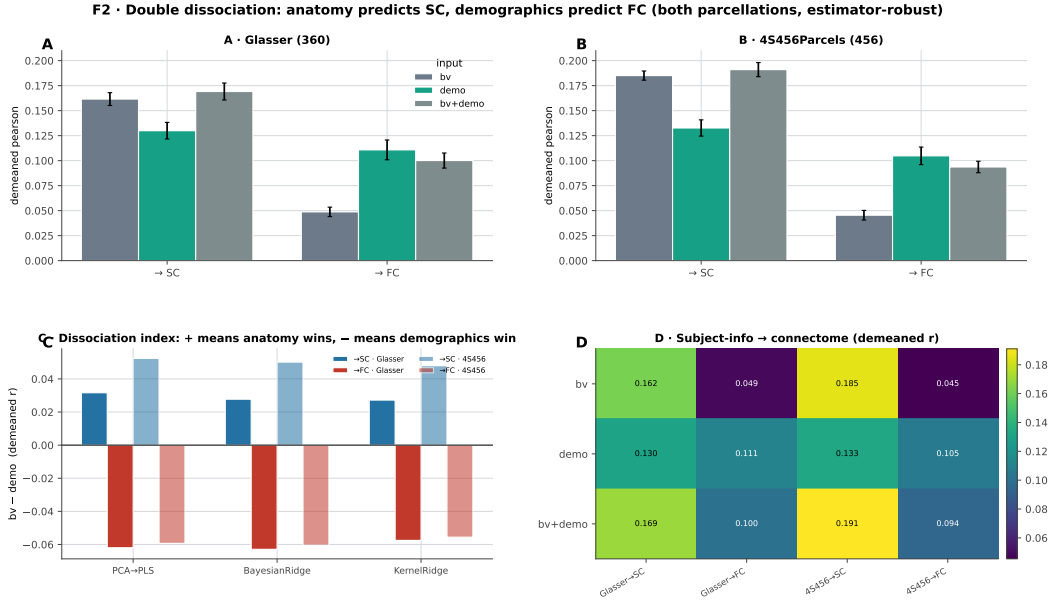


Figure 2: **F2**. (A,B) subject-information inputs → SC and FC per parcellation; (C) dissociation index across estimators; (D) full input×target×parcellation map.

3 F3 · Imputed connectomes are measurable, but their utility is asymmetric

Claim. Cross-modal prediction yields real, non-trivial imputed connectomes — but whether the imputed connectome is *useful* depends on direction, and is interpreted through F5.

Across parcellations. Both imputations are measurable: pred_SC (from FC) reaches demeaned_pearson 0.136/0.146 and pred_FC (from SC) 0.084/0.087 (Fig. 3A). Yet downstream (Fig. 3B–C) pred_SC is roughly neutral on cognition while pred_FC is harmful in both atlases — reconstruction quality and downstream utility are disconnected (a measurable connectome is not necessarily a useful one).

Status. replicated both parcellations, interpreted through F5.

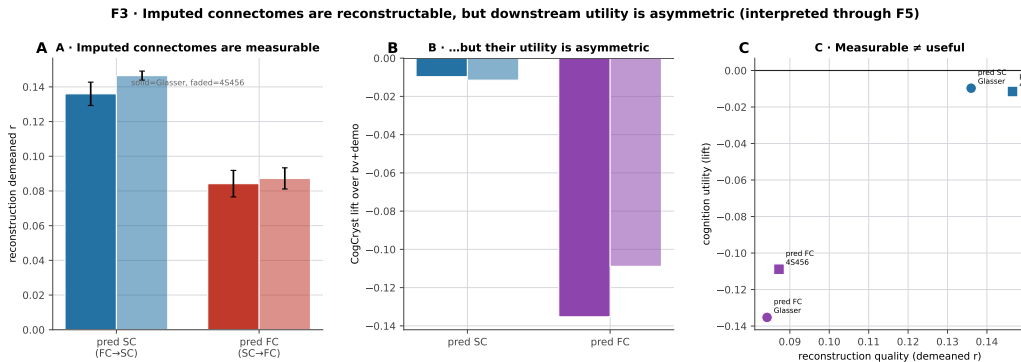


Figure 3: **F3**. (A) imputed connectomes are reconstructable; (B) but their cognition utility is asymmetric; (C) the measurable-vs-useful disconnect.

4 F4 • Observed FC carries a crystallized-cognition signal (narrow, FDR-surviving)

Claim (stated at its defensible width, not walked back later). Observed FC carries a *crystallized*-cognition signal that survives demographic residualization and restricted-family multiplicity correction in the single cell obs FC+*bv*+demo → CogCryst. The *broader* “FC beats the baseline on cognition” framing is real but multiplicity-fragile (see F11) and is *not* the headline.

The surviving cell (BayesianRidge). obs FC+*bv*+demo → CogCryst lifts +0.162 (Glasser) / +0.146 (4S456) over *bv*+demo, survives restricted-family FDR ($q = 0.027 / 0.048$, F11), and survives residualization on *bv*+demo (Fig. 4B). Plain obs FC → CogCryst (lifts +0.133 / +0.122) is nominal-only (whole-grid $q \approx 0.24$). CogTotal lifts are smaller (+0.088/+0.086) and CogFluid smaller again — the effect is carried specifically by crystallized cognition.

Across estimators. The lift is positive under PCA→PLS and BayesianRidge; KernelRidge scalar rows are shown only to expose the instability described in the estimator note (Fig. 4C). BayesianRidge is the reportable estimator.

Status. **replicated both parcellations** for the narrow cell; the broad cognition claim is multiplicity-fragile (F11). This narrow framing is consistent with F11 by construction — F4 and F11 state the same corrected claim, not a strong claim and a later retraction.

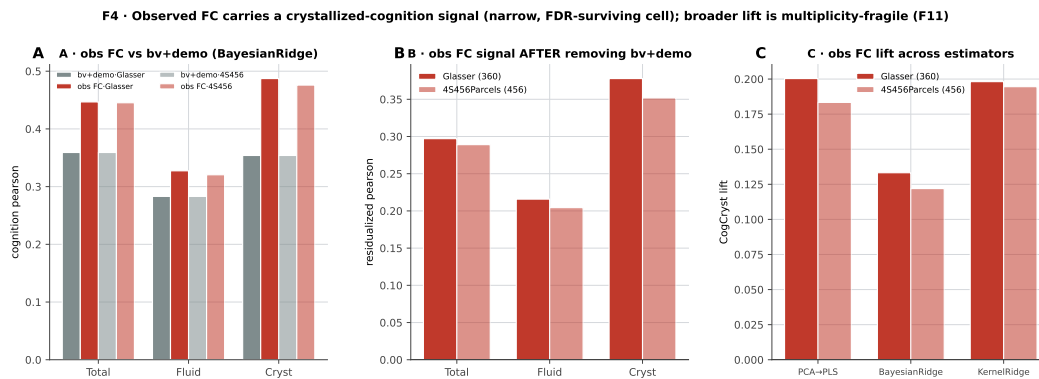


Figure 4: **F4**. (A) obs FC vs *bv*+demo for three cognitive targets; (B) FC signal after removing *bv*+demo; (C) obs FC CogCryst lift across estimators.

5 F5 • SC underperforms, imputation does not transfer utility, pred_FC is harmful

Claim. Observed SC does not clear *bv*+demo for cognition, imputed connectomes do not transfer FC’s utility, and FC imputed from SC is actively below baseline.

Across parcellations (BayesianRidge lift over *bv*+demo). Observed SC is below baseline on all three targets in both atlases (e.g. CogCryst $-0.101 / -0.095$). *pred_FC* is consistently the most harmful (CogCryst $-0.135 / -0.109$), while *pred_SC* is roughly neutral to slightly negative (Fig. 5A–B,D). The contrast *pred_SC* vs *pred_FC* at matched dimensionality shows the harm is direction-specific.

Across estimators. Under BayesianRidge (reportable) and PCA→PLS the negatives hold; the KernelRidge scalar bars flip sign (Fig. 5C) — the instability of the estimator note, not evidence SC helps.

Status. **replicated both parcellations.**

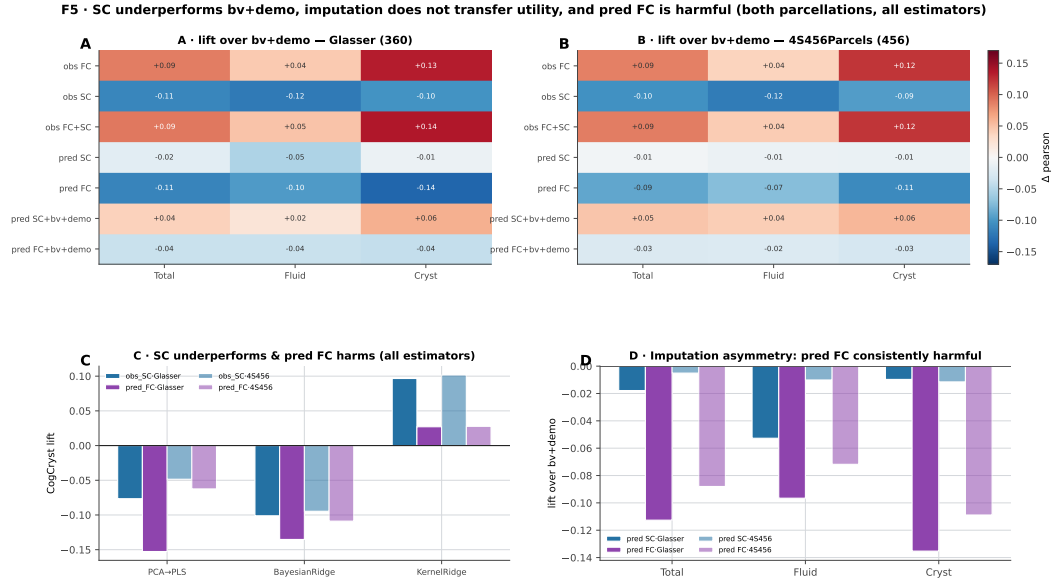


Figure 5: **F5**. (A,B) full lift-over-bv+demo maps per parcellation; (C) SC/pred_FC harm across estimators (note KR instability); (D) pred_FC harm is consistent across targets.

6 F6 · Predicted connectomes carry heritable family signal

Claim. SC predicted from FC preserves individualized, heritable structure not reducible to bv+demo.

Across parcellations. Residualized predicted SC separates siblings at $AUC \approx 0.81$ (Glasser) / 0.82 (4S456), far above the bv+demo baseline (≈ 0.56); the signal strengthens monotonically from siblings to DZ to MZ twins (Fig. 6). The sibling-AUC gain over baseline is $\approx +0.25$ in both atlases.

Source. F6–F8 come from the separate family-mechanism pipeline (reproduction/family_mechanism/outputs/), which adds the sibling/twin task on top of the same frozen family-aware splits; this task is not part of the three main grid CSVs.

Status. replicated both parcellations (family-mechanism pipeline, both parcellations).

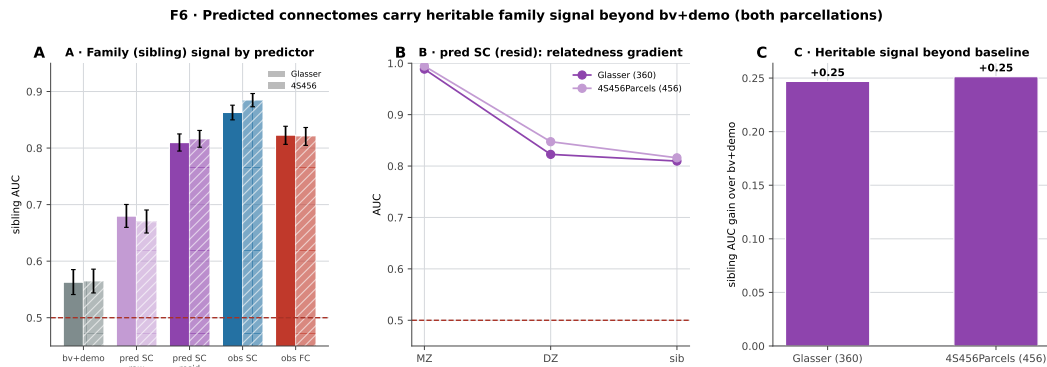


Figure 6: **F6**. (A) sibling AUC by predictor; (B) MZ/DZ/sibling gradient for residualized predicted SC; (C) AUC gain over the bv+demo baseline.

7 F7 · Reconstruction and identification objectives trade off

Claim. A predictor optimized for reconstruction does not preserve identity signal: the same imputed-SC pipeline can carry family signal or lose it depending on the objective.

Across parcellations. The reconstruction-optimized “combined” predictor keeps strong MZ/DZ separation but collapses to chance (≈ 0.50) for siblings in both atlases, whereas the residualized predictor retains sibling signal (≈ 0.81) (Fig. 7). One predicted connectome cannot serve every objective.

Status. replicated both parcellations.

F7 · Reconstruction and identification objectives trade off: the reconstruction-optimized predictor loses sibling signal (both parcellations)

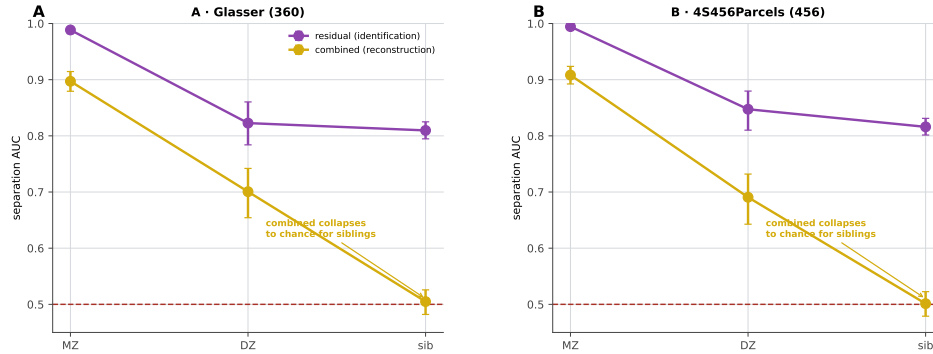


Figure 7: **F7.** Residual (identification) vs combined (reconstruction) predictors across relatedness, per parcellation; the combined predictor collapses for siblings.

8 F8 · Exploratory mechanism: an FC-predictable low-variance SC mode (component index not stable)

Claim (the claim is a *property*, not a component identity). Part of the family signal localizes to a small, FC-predictable, low-variance, family-discriminative, low-confound, visual/dorsal-attention-localized SC mode. **The component index is not a meaningful quantity** — the mode that carries this property is the 3rd PC under Glasser and the 4th under 4S456, and this migration is exactly why we make no claim about a numbered component; only the property is the finding. “PC3/PC4” below are pointers to where the property landed in each atlas, not an identity to track.

Across parcellations. The property-bearing mode (Glasser 3rd, 4S456 4th; Fig. 8A) has, among the non-PC1 components, the highest FC→PC predictability ($R^2 \approx 0.22/0.24$), sibling AUC ≈ 0.58 , and near-zero sex/volume confound. **PC1 is a confound** ($R^2 \approx 0.89\text{--}0.93$ with sex/volume; Fig. 8B) and is not a mechanism. The mode is cross-seed stable (Fig. 8C) and localizes to visual / dorsal-attention edges (Fig. 8D), a localization that survives strength+distance+reliability partialling (41% of |mode|).

Status. partial / exploratory. The property replicates across atlases; the component *label* does not, and is not claimed.

F8 · Exploratory FC-predictable low-variance SC mode (component index NOT stable: 3rd in Glasser, 4th in 4S456); PC1 is a sex/volume confound

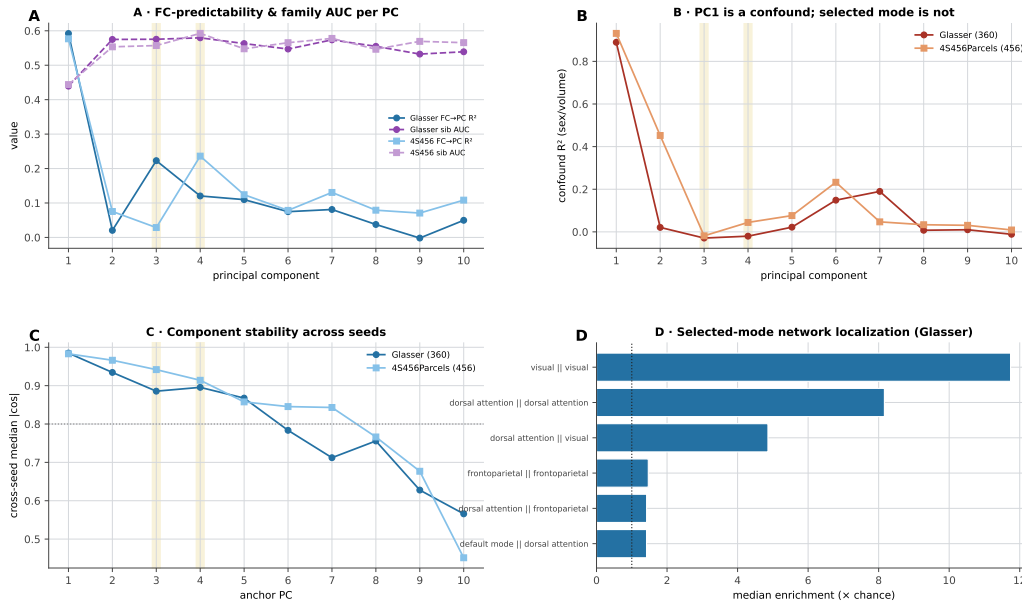


Figure 8: **F8**. (A) FC-predictability and sibling AUC per PC; (B) PC1 confound; (C) cross-seed stability; (D) network localization of the selected mode.

9 F9 · Richer tractography does not help

Claim. Named-bundle tractography (r2t) does not recover the missing structural signal that streamline counts miss.

Evidence (Glasser, exploratory). r2t predicts FC *worse* than count-SC (demeaned_pearson 0.049 vs 0.085), adds nothing marginal over counts ($\Delta = -0.001$), and carries no cognition signal above bv+demo — every tractography representation, including the r2t→synthetic-FC substitution chain, falls below the floor (Fig. 9). The FC→SC asymmetry holds across representations. Counts are a sufficient statistic for cross-modal prediction.

Status. Glasser / exploratory (the r2t bundle features were derived for Glasser).

F9 · Richer tractography (r2t) does not help: worse FC prediction, no cognition signal, asymmetry preserved (Glasser, exploratory)

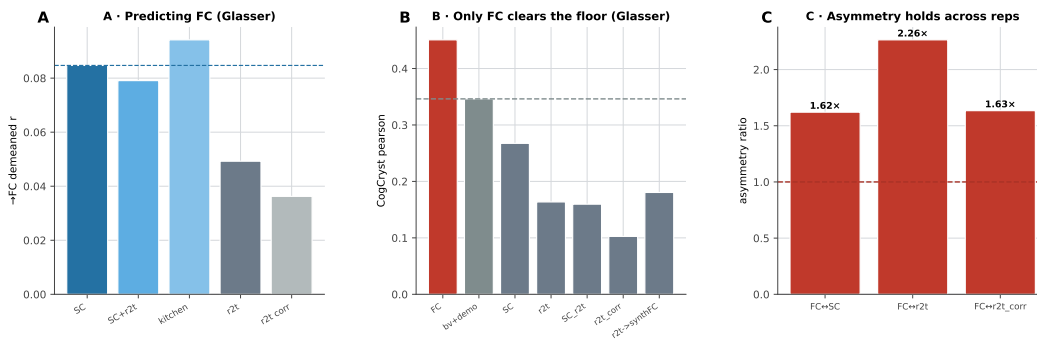


Figure 9: **F9**. (A) predicting FC from each structural representation; (B) downstream cognition by representation; (C) the asymmetry is preserved across representations.

10 F10 • Nonlinear models and more data do not break the ceiling

Claim. The linear ceiling is structural in this regime, not a model-capacity or sample-size artifact, and the SC→FC shortfall is not FC measurement noise.

Evidence. The nonlinear–linear gap stays ≤ 0 from $n = 100$ to $n \approx 683$ while the linear score climbs (Fig. 10A–B); the KernelRidge 9-variant hyperparameter sweep is nearly degenerate (within-seed spread ≈ 0.004 – 0.005 ; Fig. 10C); and SC→FC reaches only $\sim 17\%$ of the FC between-session reliability ceiling, with per-subject achieved quality uncorrelated with FC reliability (Fig. 10D). The reliability accounting is computed in both parcellations; the scaling/nonlinear probes are Glasser.

Status. Glasser / exploratory for the nonlinear+scaling probes; the reliability ceiling replicates in both parcellations.

F10 • The ceiling is structural: nonlinear models, more data, KR tuning, and FC denoising do not break it (Glasser scaling + both-parc reliability)

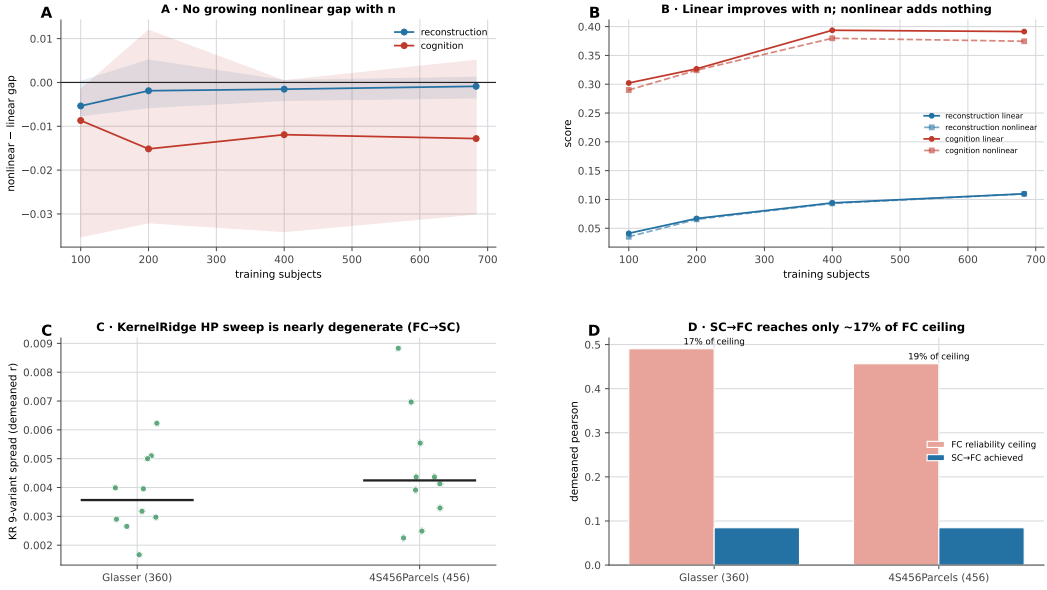


Figure 10: **F10.** (A) nonlinear gap vs n ; (B) linear improves with n , nonlinear does not; (C) KernelRidge HP flatness; (D) SC→FC vs the FC reliability ceiling in both atlases.

11 F11 • The observed-FC cognition lift is real but multiplicity-fragile

Origin. This finding is from the robustness analysis added in this work (not in the original F1–F10 ledger). It stress-tests F4.

Claim. Observed FC’s cognition lift over bv+demo is genuine but sits close to the edge of significance once multiplicity and uncertainty are handled honestly.

Evidence (both parcellations, BayesianRidge). With 95% seed CIs and BH-FDR over the downstream grid: *zero* positive lifts survive whole-grid FDR (54 cells); under a restricted affirmative family (observed-FC cognition, 12 cells) only obs FC+bv+demo → CogCryst survives (family $q = 0.027$ Glasser / 0.048 4S456), while plain obs FC → CogCryst is nominal only (uncorrected $p \approx 0.03$ – 0.04 , whole-grid $q \approx 0.24$). The median minimum-detectable-effect at 80% power is ≈ 0.056 pearson: the design is well-powered for the ~ 0.1 lifts of interest, underpowered below ~ 0.03 (Fig. 11).

Status. Both parcellations (this analysis). The affirmative claim should be reported as restricted-family-corrected with CIs, not as a bare $p < 0.05$.

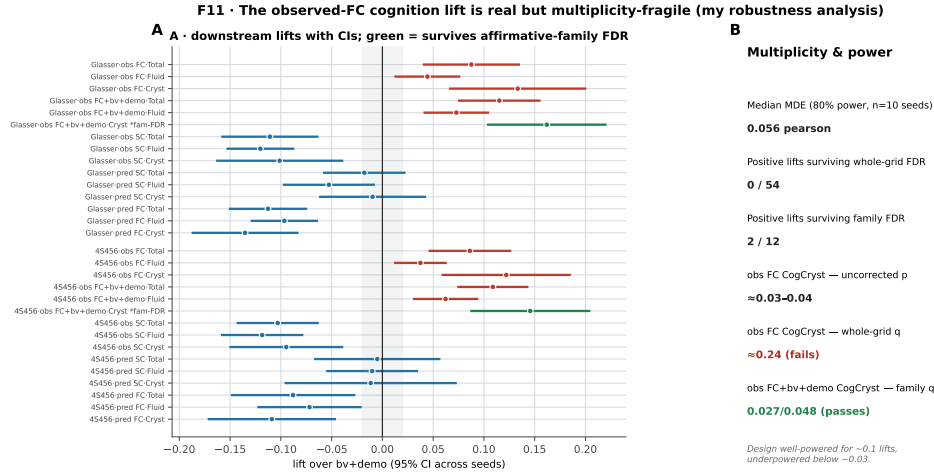


Figure 11: **F11.** (A) downstream lifts with 95% CIs; green = survives affirmative-family FDR. (B) multiplicity and power summary.

12 F12 · Bayesian corroboration (shrinkage, ROPE, Bayes factors)

Origin. A Bayesian re-analysis added in this work, corroborating F4/F5/F11 under a different inferential framework.

Claim. The frequentist conclusions hold as posterior statements: the negatives are credibly at/below a practical-equivalence region, and the positive lift is supported only for the strongest cells.

Evidence. A hierarchical partial-pooling model (PyMC NUTS) over the affirmative family shrinks each cell toward the family mean (grand-mean lift $\approx +0.10$); 11/12 cells keep a 95% HDI above the ROPE (± 0.02), with obs FC+bv+demo $\rightarrow$ CogCryst the most robust. JZS Bayes factors show strong evidence of a *harmful* effect for observed SC and pred_FC (not a null), and $P(\text{pred_FC lift} < 0) \approx 1$ (Fig. 12).

Caveat. These posteriors use the 10 seed-level lifts, so they are the Bayesian counterpart of the seed-level test (which excluded zero), not of the subject-level permutation test; because seeds are pseudo-replicated resplits, the Bayes-factor magnitudes overstate the evidence and should be read with that unit-of-analysis caveat.

Status. Both parcellations (this analysis).

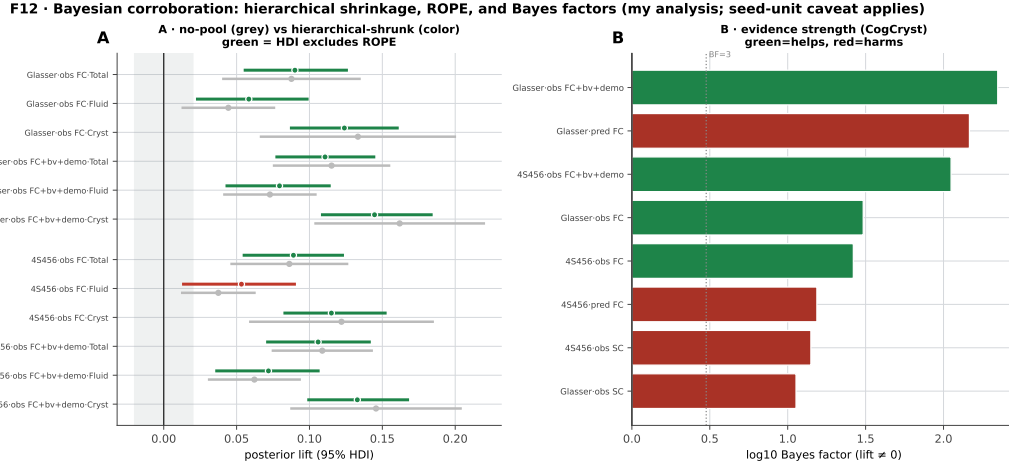


Figure 12: **F12.** (A) no-pooling vs hierarchical-shrunk posterior HDIs against the ROPE; (B) Bayes factors coloured by effect direction (green helps, red harms).

13 F13 · Imputation harm is direction-specific; cross-modal loss vs the oracle

Origin. Two quantitative results from this analysis that sharpen F3/F5 and F1/F10.

Claim 1 — direction-specific harm. `pred_FC`'s sub-baseline harm is not a generic high-dimensional-input artifact: at matched dimensionality, `pred_FC` is $\sim 2\text{--}3\times$ more harmful than `pred_SC` under BayesianRidge and PCA $\rightarrow$ PLS in both parcellations (Fig. 13A). (KernelRidge scalar bars flip sign — the instability of the estimator note, shown for transparency, not interpreted.)

Claim 2 — cross-modal loss. Using a consistent same-estimator oracle (PCA $\rightarrow$ PLS), cross-modal prediction reaches only $\approx 19\text{--}21\%$ (SC $\rightarrow$ FC / the FC $\rightarrow$ FC ceiling) and $\approx 36\text{--}38\%$ (FC $\rightarrow$ SC / the SC $\rightarrow$ SC ceiling) of the within-modality model ceiling, with credible intervals well below 1 (Fig. 13B) — the cross-modal loss is large and estimator-independent.

Status. Both parcellations (this analysis). The definitive imputation control (spectrum-matched / group-mean surrogate) needs the imputation `.npz` artifacts and is listed as a needed experiment.

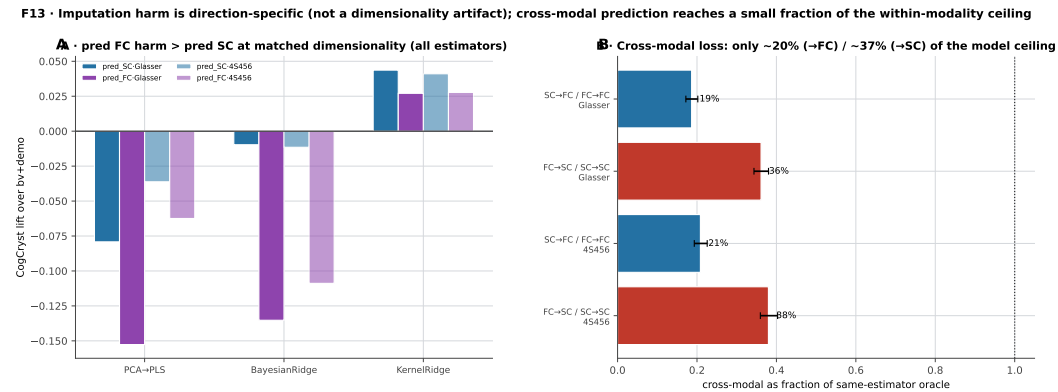


Figure 13: **F13.** (A) `pred_FC` vs `pred_SC` at matched dimensionality across estimators; (B) cross-modal prediction as a fraction of the same-estimator within-modality oracle.

Synthesis

Reading the ledger top to bottom: cross-modal prediction is *directional* (F1) and the subject-information confounds are *structured* (F2); imputed connectomes are real but their usefulness is *direction-dependent* (F3), because only observed FC carries cognition signal (F4) while SC and imputed connectomes do not clear the cheap baseline and pred_FC is harmful (F5). The constructive exception is identity: predicted connectomes carry heritable family signal (F6), but only under objectives that preserve it (F7), and the mechanism is an exploratory FC-predictable low-variance SC mode (F8). The negative results are not artifacts of representation (F9) or of model class / sample size / FC noise (F10). Across all ten, the two parcellations agree on every replicated finding; the estimator only matters for the *scalar* downstream KernelRidge rows, excluded on the principled grounds set out in the estimator note.

Three further findings from this analysis tighten the picture: the one affirmative claim (F4) is real but multiplicity-fragile (F11), and the same conclusion holds under a Bayesian framework (F12); the imputation harm is direction-specific rather than a dimensionality artifact, and cross-modal prediction reaches only a small fraction of the within-modality oracle (F13).